

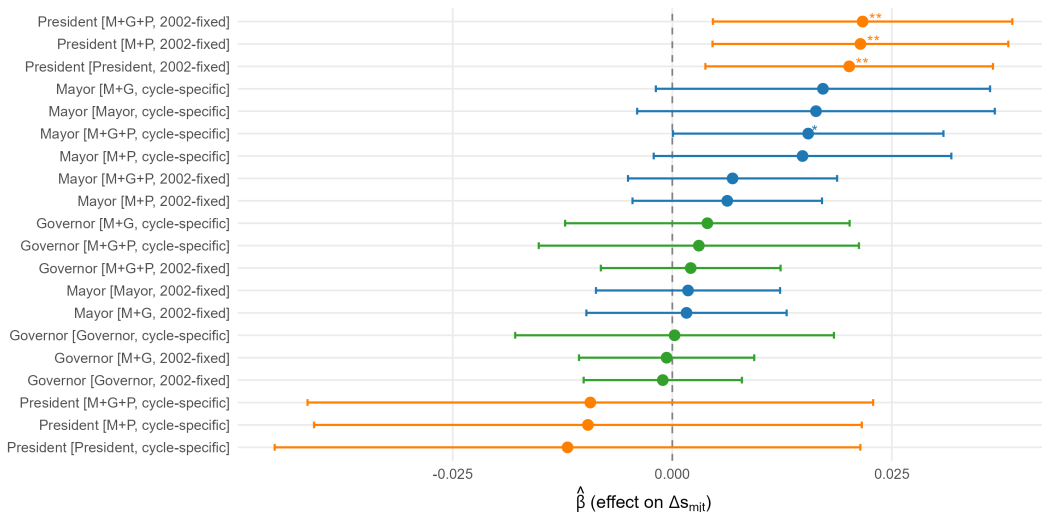
First-Stage Results — Visual Summary

February 2026

Coefficient Plot: Main Specifications

First Stage: Instrument Coefficients

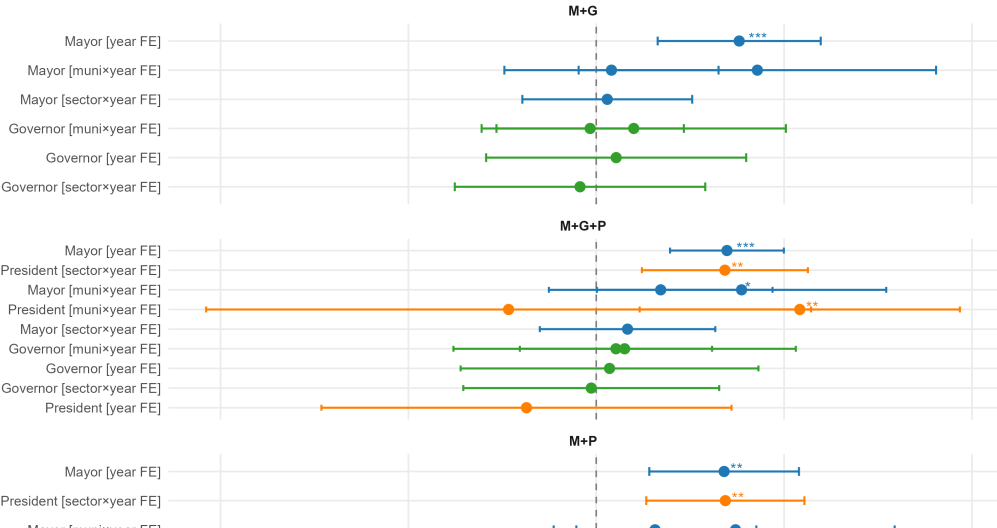
Point estimates and 95% CIs (two-way clustered SEs) | Primary FE: muni×sector + muni×year



Coefficient Plot: Robustness Across FE Specifications

First Stage: Robustness Across FE Specifications

Multi-tier instruments | Cycle-specific baseline



F-Statistic Summary

First-Stage F-Statistics Across Specifications

Wald F on excluded instruments (two-way clustered SEs) | sector_group

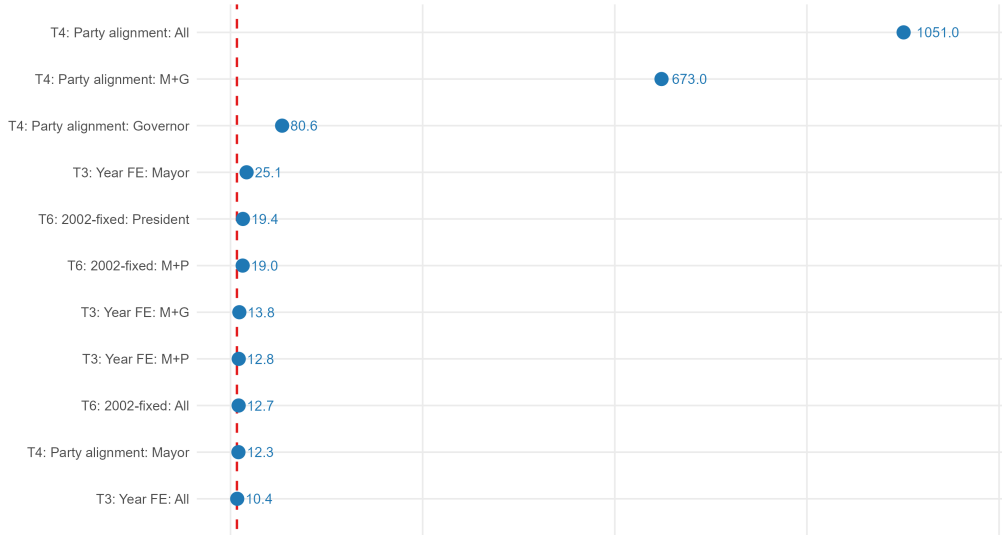


Table: Single-Tier Instruments (Coalition, Cycle-Specific)

		Δs_{mjt}	
	(1)	(2)	(3)
$Z_{coal.}^{mayor}$	0.0164 (0.0104)		
$Z_{coal.}^{gov}$		0.0003 (0.0093)	
$Z_{coal.}^{pres}$			-0.0119 (0.0170)
Observations	700,650	700,650	700,650
R ²	0.06216	0.06215	0.06216
Muni $\times$ year FE	✓	✓	✓
Muni $\times$ sector FE	✓	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table: Combined Instruments (Coalition, Cycle-Specific)

		Δs_{mjt}	
	(1)	(2)	(3)
$Z_{coal.}^{mayor}$	0.0171 (0.0097)	0.0148 (0.0086)	0.0155* (0.0079)
$Z_{coal.}^{gov}$	0.0040 (0.0083)		0.0030 (0.0093)
$Z_{coal.}^{pres}$		-0.0096 (0.0159)	-0.0093 (0.0164)
Observations	700,650	700,650	700,650
R ²	0.06216	0.06216	0.06216
Muni $\times$ year FE	✓	✓	✓
Muni $\times$ sector FE	✓	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table: Single-Tier Instruments (Coalition, 2002-Fixed)

	(1)	Δs_{mjt} (2)	(3)
$Z_{coal.}^{mayor}$	0.0018 (0.0053)		
$Z_{coal.}^{gov}$		-0.0011 (0.0046)	
$Z_{coal.}^{pres}$			0.0201** (0.0084)
Observations	700,650	700,650	700,650
R ²	0.06215	0.06215	0.06216
Muni $\times$ year FE	✓	✓	✓
Muni $\times$ sector FE	✓	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table: Combined Instruments (Coalition, 2002-Fixed)

		Δs_{mjt}	
	(1)	(2)	(3)
$Z_{coal.}^{mayor}$	0.0016 (0.0058)	0.0063 (0.0055)	0.0069 (0.0061)
$Z_{coal.}^{gov}$	-0.0006 (0.0051)		0.0021 (0.0052)
$Z_{coal.}^{pres}$		0.0214** (0.0086)	0.0217** (0.0087)
Observations	700,650	700,650	700,650
R ²	0.06215	0.06216	0.06216
Muni $\times$ year FE	✓	✓	✓
Muni $\times$ sector FE	✓	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table: Robustness — Year FE Only (Coalition)

		Δs_{mjt}	
	(1)	(2)	(3)
$Z_{coal.}^{mayor}$	0.0152*** (0.0044)	0.0136** (0.0041)	0.0139*** (0.0031)
$Z_{coal.}^{gov}$	0.0021 (0.0071)		0.0014 (0.0081)
$Z_{coal.}^{pres}$		-0.0075 (0.0105)	-0.0074 (0.0111)
Observations	700,650	700,650	700,650
R ²	0.00758	0.00758	0.00758
Year FE	✓	✓	✓
Muni × sector FE	✓	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table: Robustness — Sector×Year FE (Coalition)

		Δs_{mjt}	
	(1)	(2)	(3)
$Z_{coal.}^{mayor}$	0.0012 (0.0046)	0.0034 (0.0035)	0.0033 (0.0048)
$Z_{coal.}^{gov}$	-0.0017 (0.0068)		-0.0005 (0.0070)
$Z_{coal.}^{pres}$		0.0137** (0.0043)	0.0137** (0.0045)
Observations	700,650	700,650	700,650
R ²	0.01262	0.01262	0.01262
Sector × year FE	✓	✓	✓
Muni × sector FE	✓	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table: Exposure Control (Coalition)

	Δs_{mjt}	
	(1)	(2)
$Z_{coal.}^{mayor}$	0.0193** (0.0074)	0.0185** (0.0077)
$Z_{coal.}^{gov}$	-0.0023 (0.0122)	-0.0027 (0.0113)
Exposure control	-0.0138* (0.0065)	-0.0138* (0.0065)
$Z_{coal.}^{pres}$		-0.0043 (0.0143)
Observations	700,650	700,650
R ²	0.06238	0.06238
Muni $\times$ year FE	✓	✓
Muni $\times$ sector FE	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table: Levels Specification — Single-Tier (Coalition)

		s_mjt	
	(1)	(2)	(3)
$Z_{coal.}^{mayor,lev}$	0.2385 (0.1359)		
$Z_{coal.}^{gov,lev}$		0.2725* (0.1464)	
$Z_{coal.}^{pres,lev}$			0.3673*** (0.0915)
Observations	700,650	700,650	700,650
R ²	0.35935	0.36016	0.36013
Muni × year FE	✓	✓	✓
Muni × sector FE	✓	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table: Levels Specification — Combined (Coalition)

	s_mjt	
	(1)	(2)
$Z_{coal.}^{mayor,lev}$	0.1195 (0.0771)	0.0695 (0.0639)
$Z_{coal.}^{gov,lev}$	0.1972* (0.0985)	0.1513 (0.0879)
$Z_{coal.}^{pres,lev}$		0.2279*** (0.0256)
Observations	700,650	700,650
R ²	0.36066	0.36178
Muni × year FE	✓	✓
Muni × sector FE	✓	✓

Clustered (municipality & CNAE section) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.